

Supplementary Material A: Full Intervention Protocol

Connectivism-Informed Program for Integral AI Literacy

20-Session Schedule, Objectives, Activities, Readings, AI Tools, Assignments, and Expected Outputs

Note. This document was compiled from the facilitator manual, the standardized session protocol, and session logs maintained by the four experimental facilitators throughout the study period. All institution-identifying information has been removed for peer review. The control condition is described in Supplementary Material B.

A1. Overview and Dosage

The experimental program comprised **20 sessions of 90 minutes** each, delivered twice per week over a 10-week instructional sequence (with calendar-driven adjustments of ± 2 weeks across sections, yielding an effective range of 8–12 weeks), for a total of **30 instructional hours**. Sessions were delivered in the participants' regular classroom by four trained facilitators operating under a systematic section-rotation design.

Content covered four AI-literacy dimensions consistent with the study instrument: Conceptual Understanding (D1), Technical Skills (D2), Ethics and Social Impact (D3), and Collaboration and Knowledge Transfer (D4). Each session contributes to at least two dimensions; dimensions are indicated in the session schedule (Table A3).

Both groups (experimental and control) received the same AI fundamentals content following the same 10-week schedule and equivalent contact hours. The difference was exclusively pedagogical: the experimental group followed the connectivist framework described below; the control group received traditional instructor-led lectures (Supplementary Material B).

A2. Standard Five-Phase Session Protocol

Every session followed an identical five-phase structure regardless of topic or week. Table A1 describes each phase. The fourth phase (60–75 min) rotated across the three connectivist components on a fixed weekly schedule (Table A2).

A3. Weekly Rotation of the Ethical/Collaborative Component

A4. 20-Session Schedule

Table A3 presents the full session schedule. For each session pair (two sessions per week covering the same topic), the table specifies: topic, session objectives, learning activities (within the phases

Table A1: Standard five-phase session structure (90 min), common to all 20 sessions.

Time	Phase	Activities and facilitator role
00–10 min	Network activation	Brief review of the previous session; one trigger question linked to the knowledge-test content targeted in the session. The facilitator records student responses on a shared digital board without evaluating them, making prior knowledge visible to all teams.
10–35 min	Content node (Networked Instructional Design)	Direct presentation limited to 15 minutes maximum. Node activity: each interdisciplinary team (4–6 members) receives a differentiated case or dataset and produces a <i>knowledge node</i> —a brief structured synthesis. Teams rotate and present their node to the adjacent team (5 minutes each).
35–60 min	Technological mediation	Core hands-on activity with the designated AI tool (ChatGPT, Google Colab, or Kaggle, depending on the session). The facilitator adopts a Socratic role, circulating and posing probing questions rather than providing answers. A GitHub commit of collaborative outputs is mandatory at the end of this phase.
60–75 min	Ethical or collaborative component (weekly rotation)	See Table A2.
75–85 min	Synthesis of connections	Each team formulates one new connection that did not exist at the session’s start. The facilitator maps connections on a shared network diagram maintained across the 10 weeks.
85–90 min	Portfolio entry	Each student records individually: what was learned, how it connects with prior sessions, and one open question remaining (minimum 80 words, at least one external reference).

Table A2: Rotation of the 60–75 minute component across the 10-week sequence.

Weeks	Sessions	Component and method
1–4	1–8	Structured ethical debate. Each team analyzes a pre-assigned real-world case involving algorithmic bias, data privacy, or AI-generated content using the Structured Academic Controversy method: teams argue both sides of a dilemma before reaching a negotiated position. Facilitator mediates but does not take a stance.
5–7	9–14	Peer-teaching rotation. Students with stronger technical backgrounds act as mentors for peers from non-technical disciplines. Roles rotate across sessions to ensure bidirectional transfer. The facilitator monitors pairings and redirects where needed.
8–10	15–20	Portfolio co-evaluation. Teams apply the peer-assessment rubric to evaluate each other’s digital portfolio entries from the preceding week, providing structured written feedback. The facilitator reviews feedback quality and models constructive critique.

described in Table A1), core readings, AI tools used, assignment due, and expected output.

Note on readings. Readings listed are the primary references assigned to students as preparation or follow-up. Facilitators were provided with annotated versions of each reading. All readings were accessible through the institutional LMS.

Table A3: Full 20-session schedule. D1–D4 denote the four AI-literacy dimensions targeted; all sessions follow the five-phase protocol of Table A1.

Wk	Sess	Topic	Session objectives	Activities & AI tools	Reading & assignment	Expected output
1	1–2	What is AI? Concepts and everyday applications	(a) Distinguish AI, ML, and deep learning. (b) Identify AI applications in students’ own fields. (c) Recognize the gap between AI capabilities and common misconceptions. <i>[D1, D3]</i>	<i>Content node:</i> teams categorise 12 real-world applications as AI or non-AI, justifying each decision. <i>Tech. mediation (ChatGPT):</i> students prompt ChatGPT to explain itself, then analyze the response for accuracy and omissions. <i>Ethical debate:</i> Case 1—facial recognition in public spaces (privacy vs. security).	Ng & Soo (2024) <i>AI for Everyone</i> , Week 1 lecture notes. <i>Assignment 1:</i> portfolio entry 1 (reflection on one AI encounter in the past week).	Team knowledge-node card (categorisation matrix). Individual portfolio entry 1. GitHub repo initialised.

(Table A3 continued)

Wk	Sess	Topic	Session objectives	Activities & AI tools	Reading & assignment	Expected output
2	3–4	How do AI systems learn? From data to models	(a) Explain supervised, unsupervised, and reinforcement learning at a conceptual level. (b) Connect data quality to model behavior. (c) Use ChatGPT to explore ML concepts interactively. <i>[D1, D2]</i>	<i>Content node:</i> teams receive different toy datasets (images, text, tabular) and infer what an ML model would learn from each. <i>Tech. mediation (ChatGPT + Google Colab):</i> run a pre-built sklearn classification notebook on the Iris dataset; interpret accuracy and confusion matrix. <i>Ethical debate:</i> Case 2—predictive policing and training-data bias.	Géron (2022) <i>Hands-On ML</i> , Ch. 1 (conceptual sections only). <i>Assignment 2:</i> annotate one Colab cell block explaining what the code does in plain language.	Annotated confusion-matrix screenshot per team. Portfolio entry 2 (connection: data → model → decision).

(Table A3 continued)

Wk	Sess	Topic	Session objectives	Activities & AI tools	Reading & assignment	Expected output
3	5–6	Large language models: architecture and capabilities	(a) Describe the transformer architecture at a non-mathematical level. (b) Explain tokenization, context window, and hallucination. (c) Apply the GAME framework (Goals, Actions, Memory, Environment) to analyze ChatGPT. [D1, D2]	<i>Content node:</i> teams apply GAME to a provided ChatGPT transcript, identifying each component. <i>Tech. mediation (ChatGPT):</i> structured prompt engineering exercise—each team submits the same question with zero-shot, one-shot, and chain-of-thought prompts; compares outputs systematically. <i>Ethical debate:</i> Case 3—AI-generated academic work and institutional integrity.	Anthropic (2024 <i>AI Fluency</i> : Description and Discernment modules. <i>Assignment 3:</i> three-prompt comparison table with qualitative analysis (min. 150 words).	GAME analysis card per team. Prompt-comparison table (GitHub). Portfolio entry 3.

(Table A3 continued)

Wk	Sess	Topic	Session objectives	Activities & AI tools	Reading & assignment	Expected output
4	7–8	AI tools in practice: Google Colab and Python for data analysis	(a) Navigate the Google Colab environment independently. (b) Run and modify a basic data-analysis pipeline (load, clean, visualize). (c) Identify one disciplinary application of data analysis in the team’s fields. [D2, D4]	<i>Content node:</i> each team receives a different real-world CSV (health, education, economics) and produces a descriptive summary. <i>Tech. mediation (Google Colab + pandas):</i> teams load, clean, and produce three visualisations of their dataset; commit notebook to GitHub. <i>Ethical debate:</i> Case 4—data collection without explicit consent in research contexts.	Kaggle (2024) <i>Pandas tutorial</i> , notebooks 1–3. <i>Assignment 4:</i> individual reflection on one methodological decision made during data cleaning and its implications.	Collaborative Colab notebook per team (GitHub commit). Portfolio entry 4.

(Table A3 continued)

Wk	Sess	Topic	Session objectives	Activities & AI tools	Reading & assignment	Expected output
5	9–10	AI and bias: sources, mechanisms, and mitigation	(a) Identify three structural sources of algorithmic bias (data, model, deployment). (b) Apply a bias-evaluation lens to a real system. (c) Explain bias concepts to a peer from a different discipline. [D3, D4]	<i>Content node:</i> teams receive short case studies (hiring algorithm, medical imaging classifier, content moderation) and map bias sources using a provided scaffold. <i>Tech. mediation (ChatGPT):</i> teams prompt the model with demographic variants of the same scenario and document differential outputs. <i>Peer teaching:</i> technical-background students explain tokenization and embedding bias to social-science peers using analogies chosen collaboratively.	Barocas et al. (2023) <i>Fairness and ML</i> , Ch. 1–2 (selected sections). <i>Assignment 5:</i> two-page team brief: one real AI system, one identified bias, one proposed mitigation.	Bias-mapping worksheet per team. Differential-output log (GitHub). Portfolio entry 5.

(Table A3 continued)

Wk	Sess	Topic	Session objectives	Activities & AI tools	Reading & assignment	Expected output
6	11–12	Collaborative AI projects: Kaggle datasets and interdisciplinary problem-framing	(a) Select and justify an appropriate Kaggle dataset for a cross-disciplinary research question. (b) Formulate a problem statement combining technical and social dimensions. (c) Assign and execute team roles for a short data project. <i>[D2, D4]</i>	<i>Content node:</i> teams browse Kaggle and propose three candidate datasets; justify selection criteria (relevance, size, quality, ethical provenance). <i>Tech. mediation (Kaggle + Google Colab):</i> teams fork a beginner notebook relevant to their chosen dataset and adapt the analysis to their research question. <i>Peer teaching:</i> students with data-science backgrounds guide peers through Kaggle’s dataset documentation conventions.	Kaggle (2024) <i>Data documentation standards</i> . <i>Assignment 6:</i> one-page problem-framing document (research question, dataset, technical approach, expected societal implications).	Problem-framing document per team (LMS submission). Adapted Colab notebook (GitHub). Portfolio entry 6.

(Table A3 continued)

Wk	Sess	Topic	Session objectives	Activities & AI tools	Reading & assignment	Expected output
7	13–14	AI-generated content: risks, rights, and disciplinary implications	(a) Distinguish categories of AI-generated content (text, image, code, synthetic data). (b) Analyze copyright, attribution, and authenticity issues relevant to students' own fields. (c) Develop a personal use-policy statement for AI tools in academic work. [D3, D4]	<i>Content node:</i> teams receive a mixed set of human-written and AI-generated texts in their discipline and attempt to classify them; debrief on detection limitations. <i>Tech. mediation (ChatGPT):</i> teams use ChatGPT to draft a section of their ongoing team project, then evaluate the output against their own criteria for quality and accuracy. <i>Peer teaching:</i> students from humanities disciplines lead a discussion on attribution norms; engineering students explain code generation tools.	UNESCO (2023) <i>Guidance for Generative AI in Education</i>, Sections 2–3. <i>Assignment 7:</i> individual use-policy statement (300 words) for AI tools in their own academic program.	Classification exercise results (team). Individual use-policy statement (LMS). Portfolio entry 7.

(Table A3 continued)

Wk	Sess	Topic	Session objectives	Activities & AI tools	Reading & assignment	Expected output
8	15–16	Building and evaluating an AI project: from question to output	(a) Complete an end-to-end mini-project (data, model or prompt pipeline, output, evaluation). (b) Apply a structured rubric to evaluate a peer team’s project. (c) Provide and receive constructive written feedback. [D2, D4]	<i>Content node:</i> each team presents its project plan (5 min), receives structured oral questions from two other teams. <i>Tech. mediation (Google Colab + ChatGPT):</i> teams dedicate 25 minutes to advancing their mini-project; facilitator circulates with Socratic questions about design decisions. <i>Portfolio co-evaluation:</i> teams exchange portfolio entries from Weeks 5–7 and apply the peer-assessment rubric; written feedback returned within the session.	Team project outputs (self-generated). Peer-assessment rubric (available in the study repository alongside the facilitator manual). <i>Assignment 8:</i> revised portfolio entry 5 or 6 incorporating peer feedback.	Peer feedback forms (LMS). Revised portfolio entry (individual). GitHub commit: mini-project progress.

(Table A3 continued)

Wk	Sess	Topic	Session objectives	Activities & AI tools	Reading & assignment	Expected output
9	17–18	AI and society: labour, inequality, and access in the Global South	(a) Analyze AI’s differential societal impact across socioeconomic and geographic contexts. (b) Connect abstract ethical principles to concrete local cases (Peru/Latin America). (c) Evaluate the completeness and depth of own and peers’ ethical reasoning. [D1, D3, D4]	<i>Content node:</i> facilitator presents disaggregated data on AI adoption and digital inequality in Latin America; teams identify one local implication in their field. <i>Tech. mediation (ChatGPT):</i> teams prompt ChatGPT on a Latin American AI policy question and critically evaluate the response for regional accuracy and cultural assumptions. <i>Portfolio co-evaluation:</i> teams evaluate portfolio entries 6–7 from a peer team, focusing on depth of ethical reasoning using the rubric.	Valdivieso (2025) on AI access gaps in Latin America (selected sections). Torres-Díaz et al. (2025) on AI knowledge and academic integrity. <i>Assignment 9:</i> team position paper (400 words): one AI policy recommendation for the institution, with ethical justification.	ChatGPT response analysis (team, GitHub). Position paper (LMS submission). Portfolio entry 9.

(Table A3 continued)

Wk	Sess	Topic	Session objectives	Activities & AI tools	Reading & assignment	Expected output
10	19–20	Integration and transfer: what I know, what I can do, what remains open	(a) Synthesise learning across the four AI-literacy dimensions. (b) Identify transferable competencies to other courses or professional contexts. (c) Formulate three specific open questions for continued learning. <i>[D1, D2, D3, D4]</i>	<i>Content node:</i> each team maps its collective learning journey on a 10-week network diagram, identifying the strongest and weakest connections. <i>Tech. mediation (ChatGPT + Colab):</i> teams complete their mini-project; final GitHub commit and Colab notebook submitted. <i>Portfolio co-evaluation:</i> full portfolio (entries 1–9) submitted to peer team for final holistic evaluation using complete rubric; written synthesis feedback returned.	No new external reading. <i>Assignment 10 (summative):</i> final individual portfolio (entries 1–9 plus a 500-word integrative reflection); final team mini-project report (submitted via LMS and GitHub).	Final digital portfolio per student (LMS). Final mini-project report per team (LMS + GitHub). Completed network diagram per team.

A5. AI Tools: Purposes and Access

A6. Assessment and Expected Outputs

A7. Allocation and Measurement Timeline

Assignment of the eight intact sections to conditions was completed prior to baseline data collection, following the institutional academic timetable established before student enrolment. Students were not informed of their section’s condition until the first session of Week 1.

- **Week 0 (T0).** Both groups completed the pretest (Self-Perception Scale, Objective Knowledge

Table A4: AI tools used in the experimental condition, their primary purposes, and the sessions in which they were employed.

Tool	Primary purpose	Dual role in the framework	Sessions
ChatGPT	Prompt engineering, concept exploration, text generation and critique	Productivity aid <i>and</i> object of study (GAME analysis, bias probing, accuracy evaluation)	1–2, 3–4, 5–6, 7–8, 9–10, 13–14, 15–16, 17–18, 19–20
Google Colab	Collaborative Python programming, data analysis, model demonstrations	Shared coding environment enabling real-time interdisciplinary collaboration	3–4, 7–8, 11–12, 15–16, 19–20
Kaggle	Real-world dataset access and beginner notebook repository	Contextualised data source grounding technical exercises in societal problems	11–12, 15–16, 19–20
GitHub	Version control and collaborative code repository	Mandatory commit each session; provides traceable evidence of iterative knowledge construction	All sessions (commits mandatory)
Institutional LMS	Material distribution, discussion forums, assignment submission	Central hub for Networked Instructional Design; persistent record of portfolio and peer feedback	All sessions

Table A5: Assessment components, their format, weight, and connection to AI-literacy dimensions.

Component	Format	Weight	Dimensions	Description
Digital portfolio (individual)	9 weekly entries + 1 integrative reflection	40%	D1, D2, D3, D4	Each entry: ≥ 80 words, ≥ 1 external reference, explicit connection to prior session. Assessed with the portfolio rubric (available in the study repository alongside the facilitator manual).
Team mini-project	Colab notebook + written report (400–600 words)	35%	D2, D4	End-to-end AI project on a Kaggle dataset: problem framing, data pipeline, output, evaluation, and one societal implication. GitHub repository required.
Peer-assessment participation	Written feedback forms (3 rounds)	15%	D4	Rounds in Weeks 8, 9, and 10. Assessed on specificity, constructiveness, and alignment with rubric criteria.
Weekly knowledge-node cards	Team artefact (one per session pair)	10%	D1, D2	Brief structured synthesis produced during the content-node phase; submitted via LMS. Graded for accuracy and conceptual coherence.

Test, and Demographic Questionnaire) online under standardized conditions ($\approx 25\text{--}30$ minutes), before any differentiated activity.

- **Weeks 1–10 (Intervention).** Experimental group followed the connectivist program; control group received traditional instruction on the same AI content.
- **Weeks 8–12 (T1, posttest).** Administered in the final week of the intervention in each section (variation due to academic calendar; range 8–12 weeks).
- **T1 + 4 weeks (T2, follow-up).** Four weeks after the conclusion of the intervention in each section; Parts A and B only.

Upon completion of the study, participants in the control group were offered full access to the connectivist framework materials, including the collaborative activities, portfolio templates, and digital resources used in the experimental condition (waitlist strategy).